

QUANTITATIVE TRADING

WITH MACHINE LEARNING

Problem #43 • Paper-inspired supply-chain forecasting

Predicting Volkswagen (VWAGY) returns using market features + supply-chain proxies

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FINAL REVIEW & DEMO

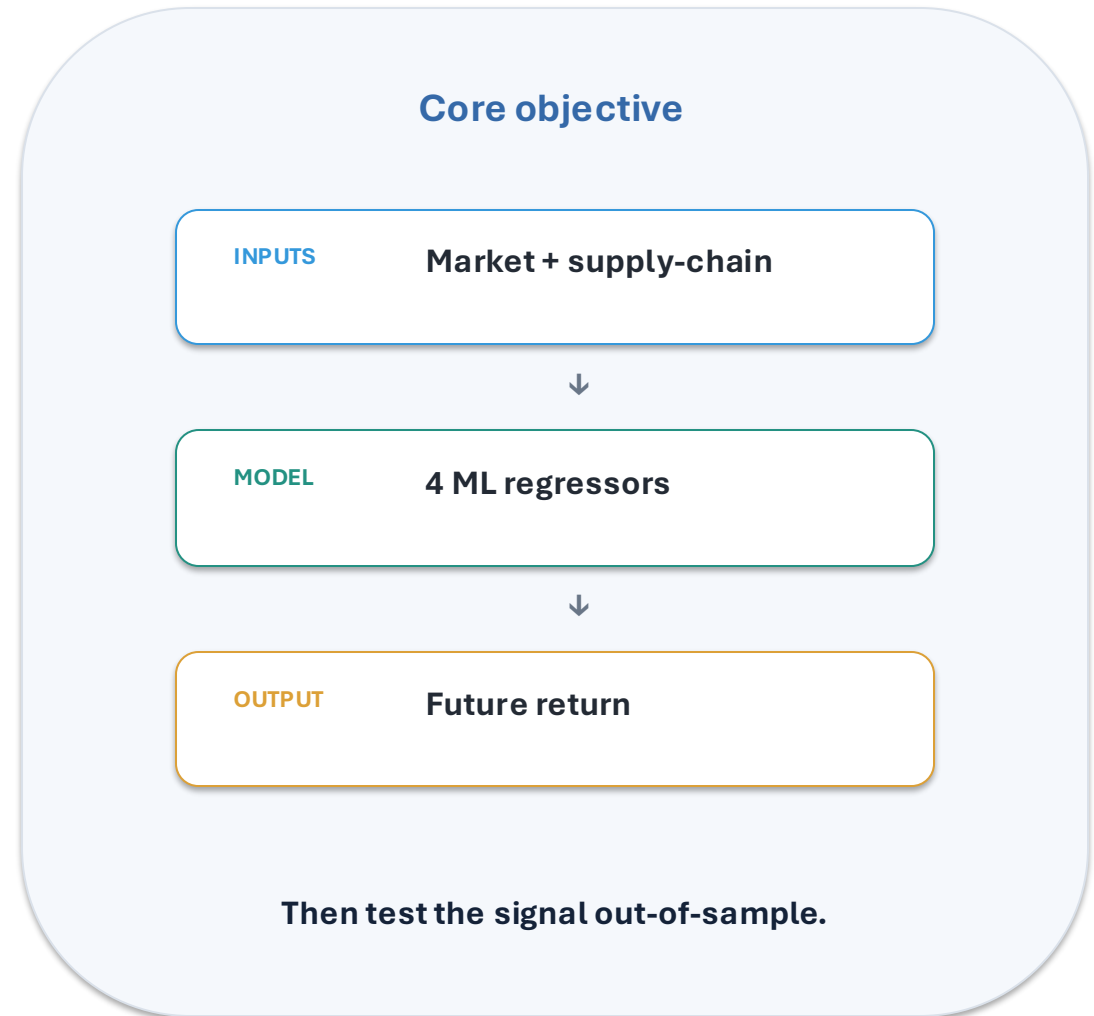
1. Problem & Motivation

Research question

Can machine learning extract a useful signal for future VWAGY returns from historical VWAGY behaviour and related supply-chain companies?

Why this is difficult

- Short-term stock returns are noisy.
- Relationships change across market regimes.
- Random train/test splits can leak future regimes.
- A profitable-looking period is not enough; unseen periods matter.



2. Dataset & Supply-Chain Proxies

TARGET

VWAGY

Volkswagen AG

MARKET PERIOD

2015–2025

Yahoo Finance daily data

PROXIES

9

Public supply-chain proxies

FEATURES

276

24 technical + 252 proxy lags

Proxy universe

- CON.DE Continental
- IFX.DE Infineon
- HLE.DE HELLA
- BAS.DE BASF
- BWA BorgWarner
- MGA Magna
- LEA Lear
- APTV Aptiv
- TSM TSMC

Feature construction

9 proxies × 28 return lags

= 252 supply-chain features

+ 24 VWAGY technical features

= 276 total features

3. Feature Engineering

FEATURES

Returns

3, 5, 10, 20, 30, 60-day returns

Volatility

Rolling volatility at the same windows

Absolute return

Magnitude of recent movement

Momentum

Price relative to rolling mean

SUPPLY-CHAIN SIGNAL

For each proxy: historical daily returns lagged from $t-1$ through $t-28$

This captures whether movements in related firms precede VWAGY movements.

Targets: future VWAGY return at 1, 5, 10, 15, 20 and 25 trading days.

4. Leakage-Safe Rolling Evaluation



Every rolling window

- Evaluate all 4 models × all 6 horizons on validation RMSE.
- Select the best configuration using validation only.
- Retrain selected model on train + validation.
- Predict the next month as unseen out-of-sample data.

Leakage protections

- Chronological split — no random shuffling
- Future prices used only as labels
- Test month never used for selection
- Elastic Net scaling inside training pipeline

5. Models & Rolling Selection

Elastic Net

Regularized linear model

27

windows selected

XGBoost

Gradient-boosted trees

18

windows selected

Decision Tree

Nonlinear tree model

8

windows selected

LightGBM

Efficient boosting model

2

windows selected

Forecast-horizon selection

1D • 5D • 10D • 15D • 20D • 25D

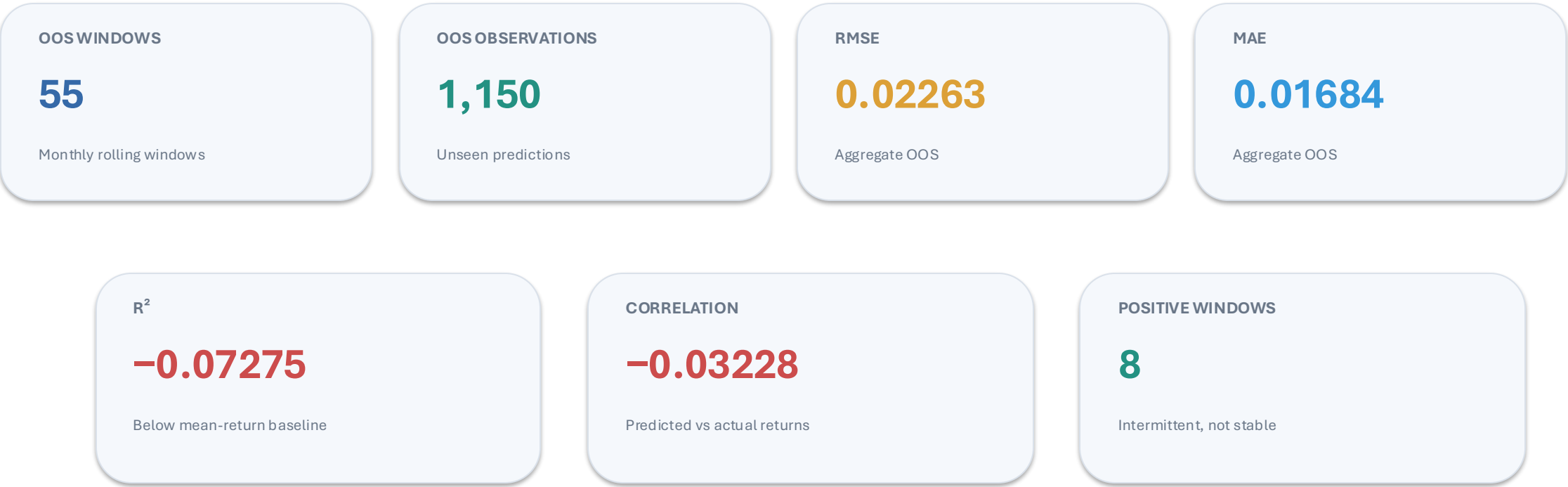
Selected horizon in all 55 windows: 1 day

What this tells us

No single model dominated every market period.

Model choice is regime-dependent.

6. Out-of-Sample Results



Interpretation: individual market regimes produced useful-looking windows, but the aggregate OOS signal was not stable enough to beat the simple mean-return baseline.

This is a valid negative research result — the methodology is tested rather than assumed to work.

7. Trading Strategy & Benchmark

Signal rule

Prediction > +threshold → LONG

Prediction < -threshold → SHORT

Otherwise → NEUTRAL

Tested thresholds

0.00% • 0.25% • 0.50% • 1.00%

Best ML strategy vs Buy & Hold

Metric	ML 0.50%	Buy & Hold
Total return	-13.05%	-70.39%
Sharpe	-0.177	-0.675
Max drawdown	-28.63%	-77.52%

Best by Sharpe: 0.50% threshold

Even the best ML threshold remained negative in aggregate.

The benchmark also had a difficult period; the ML signal reduced drawdown but did not produce positive aggregate return.

8. What Did We Learn?

Main findings

- Supply-chain proxy information can be engineered into a high-dimensional time-series dataset.
- Model performance changes across market regimes.
- The 1-day horizon was consistently preferred by validation.
- Positive windows were intermittent, not stable.
- Aggregate OOS R^2 was negative.
- Trading performance did not establish a robust profitable edge.

Why might this happen?

- Financial returns contain substantial noise.
- Market relationships are regime-dependent.
- Only a subset of the broader supplier network is available publicly.
- Macro variables from the reference methodology are not included.
- Correlated lag features can increase complexity and overfitting risk.

9. Live Demonstration Plan

01

Data

Run collection cell → show VWAGY + 9 proxies

02

Features

Show 276 features → 24 technical + 252 supply-chain

03

Rolling evaluation

Run model selection → show 55 windows / 1,150 OOS

04

Results

Show aggregate RMSE / MAE / R^2 + model frequency

05

Trading

Show threshold signals + equity curve

06

Conclusion

Compare best ML strategy with Buy & Hold

Demo principle: show the pipeline running — don't just show screenshots.

10. Conclusion

The project builds a complete, leakage-aware quantitative ML pipeline — from market data to features, model selection, unseen evaluation and trading signals.

ENGINEERING

276 features • 4 models • rolling OOS evaluation

RESULT

Negative aggregate predictive signal • honest out-of-sample finding

NEXT STEP

Broader suppliers • macro variables • stronger validation • realistic costs

Thank you

Questions & discussion